Book Introduction:
Traffic Flow Dynamics Data, Models and Simulation

Chapter 3&5

Chapter 3 :
Cross-Sectional Data

Cross-Sectional Data

The resource of Cross-sectional data

- Cross-sectional data is captured by **stationary induction loops**, **radar**, or **infrared sensors**.

Data Format

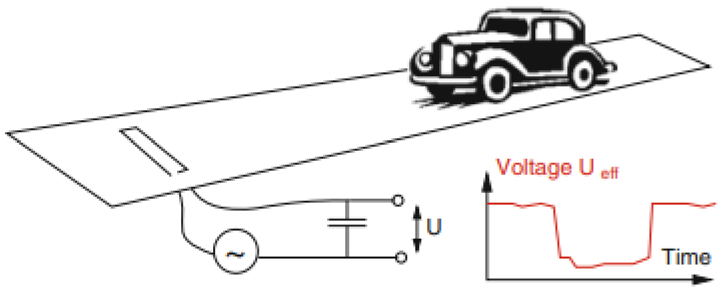
- Either directly as **single vehicle data** or aggregated into **macroscopic quantities**.

In this chapter we are going to :

1. Define the measurable and derived quantities characterizing both data formats.
2. Attention the difference between **temporal and spatial averages**.
3. Estimate **traffic density**.
4. Introduce speed estimation methods to overcome the inability of **single-loop detectors** to directly measure vehicle speed.

Cross-Sectional Data

Most commonly, **induction loops** are installed beneath the road surface. A **single-loop detector** can directly measure (only) the following quantities:



t_α^0 : The front of vehicle α passes the detector (voltage drop)

t_α^1 : The rear end of vehicle α passes the detector (voltage drop)

We can obtain an estimate in the case of relatively uniform speed values by assuming an average vehicle length l (But it will make a large errors)

Cross-Sectional Data

The value which **double-loop detector** detected and the corresponding **microscopic quantities**

- Length of the vehicle $\boldsymbol{l_\alpha}$
- Based on l_α , the **vehicle type** can be estimated
- **Time headway** between the front bumpers of successive vehicles (the smaller index $\alpha - 1$ denotes the leading vehicle)

$$l_\alpha = v_\alpha (t_\alpha^1 - t_\alpha^0)$$

- Time gap between the rear and front bumpers

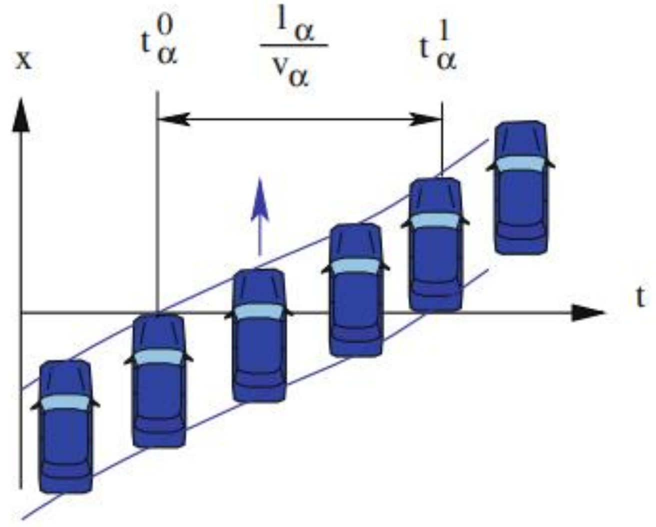
$$\Delta t_\alpha = t_\alpha^0 - t_{\alpha-1}^0$$

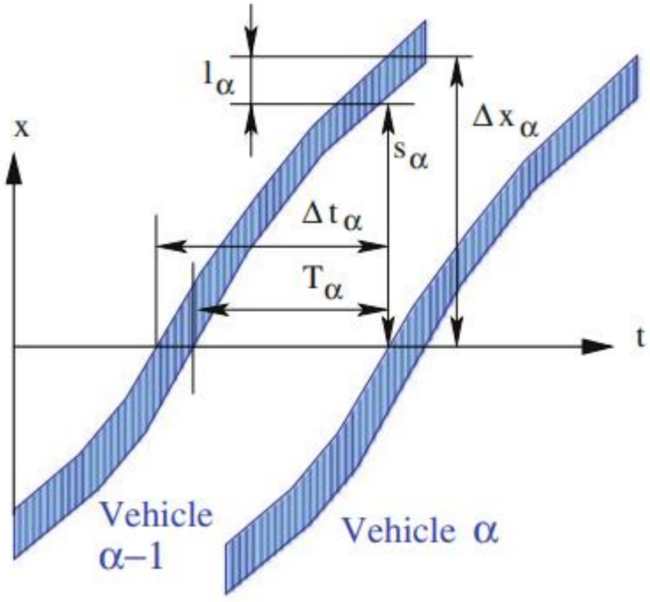
- Distance headway

$$d_\alpha = v_{\alpha-1} \Delta t_\alpha$$

- Distance gap between the rear and front bumpers

$$s_\alpha = d_\alpha - l_{\alpha-1}$$





Aggregated Data

Most detectors **aggregate** the microscopic single-vehicle data by averaging over **fixed time intervals** Δt and transmit only the macroscopic data (aggregated data) to the traffic control center.

- Save bandwidth in the transmission and disk space
- All the microscopic information is lost.
- The most common being $\Delta t = 60$ s

Averages over a fixed number of vehicles (e.g. $\Delta N = 50$ veh) are rarely used, even though they are statistically more meaningful.

Definition of **Traffic Flow**

- The traffic flow is defined as the **number of vehicles** ΔN passing the cross-section at location x within a time interval Δt

$$Q(x,t) = \frac{\Delta N}{\Delta t}$$

It is usually given in units of vehicles per hour (*veh/h*) or vehicles per minute.

Aggregated Data

$$Q(x,t) = \frac{\Delta N}{\Delta t}$$

- In terms of the microscopic quantities, the traffic flow Q can be considered as the inverse of the time mean of the headways, i.e.

$$Q = \frac{1}{<\Delta t_\alpha>}$$

- Inverse of the headway is called **microscopic flow**

$$q_\alpha = \frac{1}{< \Delta t_\alpha >}$$

- We emphasize that the traffic flow Q can be considered as the harmonic mean of the microscopic flow

$$Q = \frac{1}{\langle \Delta t_\alpha \rangle} = \frac{1}{\langle 1/q_\alpha \rangle}$$

Occupancy

- The dimensionless occupancy is the fraction of the aggregation interval during which the cross-section is occupied by a vehicle

$$O(x,t) = \frac{1}{\Delta t} \sum_{\alpha=\alpha_0}^{\alpha_0+\Delta N-1} (t_\alpha^1 - t_\alpha^0)$$

Aggregated Data-Velocity estimation

Arithmetic mean speed : The arithmetic mean speed is the average speed of the ΔN vehicles passing the cross-section during the aggregation interval

$$V(x,t) = \langle v_\alpha \rangle = \frac{1}{\Delta N} \sum_{\alpha=\alpha_0}^{\alpha_0+\Delta N-1} v_\alpha .$$

Harmonic mean speed (Space mean speed):

$$V_{\mathrm{H}}(x,t) = \frac{1}{\left\langle \frac{1}{v_\alpha} \right\rangle} = \frac{\Delta N}{\sum_{\alpha=\alpha_0}^{\alpha_0+\Delta N-1} \frac{1}{v_\alpha}}$$

When neglecting accelerations, VH corresponds approximatively to the (spatial) average of the speed at a fixed time instant.

Always $V_{\mathrm{H}} \le V$

Arithmetic time mean of microscopic flow. The arithmetic time mean of microscopic flow is defined by

$$Q^*(x,t) = \langle q_\alpha \rangle = \left\langle \frac{1}{\Delta t_\alpha} \right\rangle = \frac{1}{\Delta N} \sum_{\alpha=\alpha_0}^{\alpha_0+\Delta N-1} \frac{1}{\Delta t_\alpha}$$

The **speed variance** is a measure of the spread of the speed values within the aggregation interval.

$$\mathrm{Var}(v) = \sigma_v^2(x,t) = \langle (v_\alpha - \langle v_\alpha \rangle)^2 \rangle = \langle v_\alpha^2 \rangle - \langle v_\alpha \rangle^2$$

Estimating Spatial Quantities from Cross-Sectional Data

The traffic density is defined as a spatial average at a fixed time (the number of vehicles on a given road segment) but cross-sectional detectors can only measure temporal averages at a fixed location (the cross-section)

$$\rho(x,t) = \frac{Q(x,t)}{V(x,t)} = \frac{\text{flow}}{\text{speed}}$$

- This equation implicitly assumes that the speed V is a spatial average.
- Using the temporal averages obtained from cross-sectional detectors induces systematic errors.
- Faster vehicles are "seen" more frequently by detectors than slower vehicles, yielding a bias towards larger speed values.

Estimating Spatial Quantities from Cross-Sectional Data

We can obtain a better estimate for the density from its definition "vehicles per distance", which can be expressed in terms of microscopic quantities as the inverse of the space mean of the distance headways

$$\rho(x,t) = \frac{1}{\langle d_\alpha \rangle} = \frac{\Delta N}{\sum_\alpha d_\alpha}$$

For a given fixed time interval,

$$\Delta t = \sum_{\alpha=\alpha_0}^{\alpha_0+\Delta N-1} \Delta t_\alpha = \Delta N \langle \Delta t_\alpha \rangle$$

The flow Q ("vehicles per time") can be written as the inverse of the time mean of the headways

$$Q = \frac{\Delta N}{\Delta t} = \frac{1}{\langle \Delta t_\alpha \rangle}$$

We will discuss two different ways for expressing the density in terms of the measurable quantities Δt_α and v_α

Estimating Spatial Quantities from Cross-Sectional Data

$$\begin{cases} d_\alpha = v_{\alpha-1}\Delta t_\alpha \\ \rho(x,t) = \dfrac{1}{\langle d_\alpha \rangle} = \dfrac{\Delta N}{\sum_\alpha d_\alpha} \\ Q = \dfrac{\Delta N}{\Delta t} = \dfrac{1}{\langle \Delta t_\alpha \rangle} \end{cases}$$

$$\begin{aligned} \frac{1}{\rho} &= \langle d_\alpha \rangle = \langle v_{\alpha-1}\Delta t_\alpha \rangle \\ &\approx \langle v_\alpha \Delta t_\alpha \rangle \\ &= \langle v_\alpha \rangle \langle \Delta t_\alpha \rangle + \mathrm{Cov}(v_\alpha, \Delta t_\alpha) \\ &= \frac{V}{Q} + \mathrm{Cov}(v_\alpha, \Delta t_\alpha), \end{aligned}$$

$$\rho = \frac{Q}{V}\left(\frac{1}{1 + \frac{Q}{V}\mathrm{Cov}(v_\alpha, \Delta t_\alpha)}\right)$$

The covariance is positive if both variables are positively correlated, i.e. larger values of x tend to be accompanied by proportionally larger values of y. The significance of such a linear relationship is quantified by the correlation coefficient

$$r_{x,y} = \frac{\mathrm{Cov}(x, y)}{\sigma_x \sigma_y}$$

The correlation coefficient can help us get Wardrop's equation

$$\rho = \frac{Q}{V}\left(\frac{1}{1 + \frac{\sigma_V}{V}\frac{Q}{\sigma_Q} r_{v_\alpha, \Delta t_\alpha}}\right)$$

The relation Q/V systematically underestimates the real density in congested traffic

Estimating Spatial Quantities from Cross-Sectional Data

$$\begin{cases} d_\alpha = v_{\alpha-1}\Delta t_\alpha \\ \rho(x,t) = \dfrac{1}{\langle d_\alpha \rangle} \\ Q = \dfrac{\Delta N}{\Delta t} = \dfrac{1}{\langle \Delta t_\alpha \rangle} \end{cases}$$

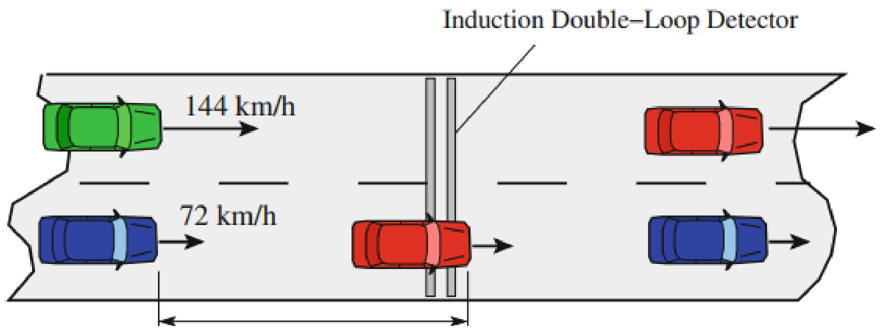
Q/VH generally overestimates the real density

$$\begin{aligned} \frac{1}{Q} &= \langle \Delta t_\alpha \rangle = \left\langle \frac{d_\alpha}{v_{\alpha-1}} \right\rangle \\ &\approx \left\langle \frac{d_\alpha}{v_\alpha} \right\rangle = \langle d_\alpha \rangle \left\langle \frac{1}{v_\alpha} \right\rangle + \operatorname{Cov}\left(d_\alpha, \frac{1}{v_\alpha}\right) \\ &= \frac{1}{\rho V_{\mathrm{H}}} + \operatorname{Cov}\left(d_\alpha, \frac{1}{v_\alpha}\right). \end{aligned}$$

$$\rho = \frac{Q}{V_{\mathrm{H}}}\left(\frac{1}{1 - Q\operatorname{Cov}(d_\alpha, 1/v_\alpha)}\right)$$

Discussion of the Two Approximations

In practice, the covariances in above 2 formula are usually assumed to be zero



$$\rho^{(1)} = \frac{Q}{V} \quad \text{or} \quad \rho^{(2)} = \frac{Q}{V_{\mathrm{H}}}$$

1. If all vehicle speeds v_α are the same, then $V = V_H$ and thus $\rho \ = \rho^{(1)} = \rho^{(2)}$.

$$\rho = \frac{Q}{V}\left(\frac{1}{1 + \frac{\sigma_V}{V}\frac{Q}{\sigma_Q} r_{v_\alpha, \Delta t_\alpha}}\right)$$

1. If all headways Δt_α are the same, then $Cov(v_\alpha, \Delta t_\alpha) = 0$ and thus $\rho = \rho^{(1)} = \frac{Q}{V}$ holds exactly. Otherwise, $\rho^{(1)}$ most likely underestimates the real density as $Cov(v_\alpha, \Delta t_\alpha)$ is usually negative.

2. If all distance headways d_α are the same, then $Cov(d_\alpha, \frac{1}{v_\alpha}) = 0$ and thus $\rho \ = \rho^{(2)} = \frac{Q}{V_H}$ holds exactly. Otherwise, $\rho^{(2)}$ most likely overestimates the real density since $Cov(d_\alpha, \frac{1}{v_\alpha})$ is usually negative as well.

$$\rho = \frac{Q}{V_{\mathrm{H}}}\left(\frac{1}{1 - Q \operatorname{Cov}(d_\alpha, 1/v_\alpha)}\right)$$

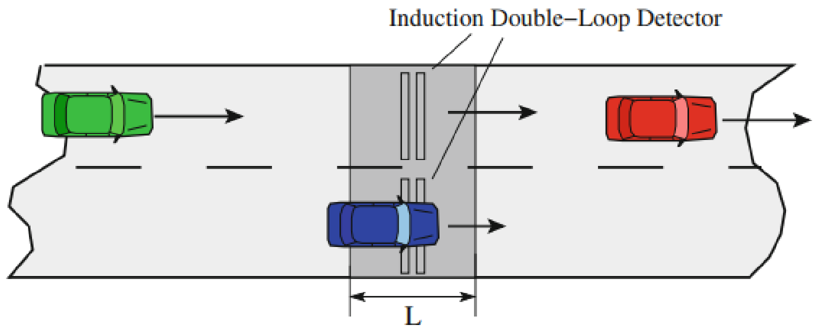
Space Mean Speed

Defination : The space mean speed (instantaneous mean) $V(t)$ is the arithmetic mean of the speed of all vehicles within a given road segment at time t

$$\langle V(t) \rangle = \frac{1}{n(t)} \sum_{\alpha=1}^{n(t)} v_\alpha(t)$$

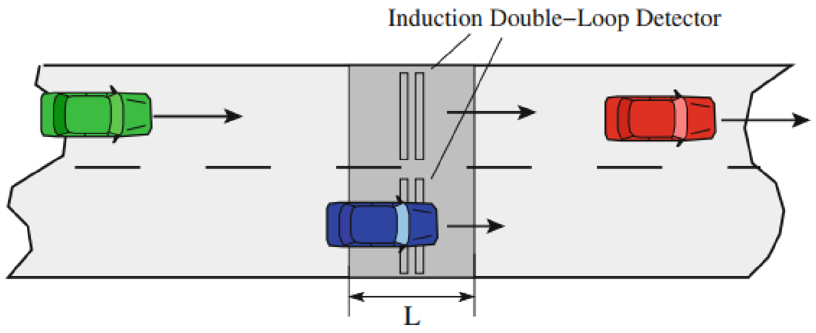
- The number and identities of vehicles used in the average in equation changes within the aggregation interval Δt.

- It is possible that n(t) = 0



Space Mean Speed

Defination : The space mean speed (instantaneous mean) $V(t)$ is the arithmetic mean of the speed of all vehicles within a given road segment at time t



$$\langle V(t)\rangle = \frac{1}{n(t)}\sum_{\alpha=1}^{n(t)} v_\alpha(t)$$

Get a more suitable definition by averaging the instantaneous mean over the aggregation interval Δt

$$\langle V\rangle = \frac{\int_t^{t+\Delta t} n(t')\langle V(t')\rangle\,\mathrm{d}t'}{\int_t^{t+\Delta t} n(t')\,\mathrm{d}t'} = \frac{\sum_\alpha \int_{t_\alpha}^{t_\alpha+\tau_\alpha} v_\alpha(t')\,\mathrm{d}t'}{\sum_\alpha \tau_\alpha}$$

$$\approx \frac{\sum_\alpha \tau_\alpha v_\alpha}{\sum_\alpha \tau_\alpha} \approx \frac{n\,L}{\sum_\alpha L/v_\alpha}$$

$$= \frac{n}{\sum_{\alpha=1}^{n} \frac{1}{v_\alpha}},$$

Let the reference length L small enough so that no vehicles are on the reference road segment at time t or $t + \Delta t$

Assume vehicle speed does not change significantly during the time needed for passing the segment, $\tau_\alpha \approx \mathrm{L}/\, v_\alpha$

Determining Speed from Single-Loop Detectors

On actual roadways, single-loop detectors are the least expensive and most widely deployed. Single-loop detectors only measure the entry and exit times t_α^0 and t_α^1 of each vehicle α.

Although we do not know each vehicle's length, we can assume an average vehicle length $< l_\alpha >$

$$O(x,t) = \frac{1}{\Delta t} \sum_{\alpha=\alpha_0}^{\alpha_0+\Delta N-1} (t_\alpha^1 - t_\alpha^0).$$

$$\begin{aligned} O &= \frac{1}{\Delta t} \sum_\alpha (t_\alpha^1 - t_\alpha^0) \\ &= \frac{1}{\Delta t} \sum_\alpha \frac{l_\alpha}{v_\alpha} \\ &= \frac{n}{\Delta t} \left[\langle l_\alpha \rangle \left\langle \frac{1}{v_\alpha} \right\rangle + \mathrm{Cov}\left(l_\alpha, \frac{1}{v_\alpha} \right) \right] \\ &= Q \left[\langle l_\alpha \rangle \left\langle \frac{1}{v_\alpha} \right\rangle + \mathrm{Cov}\left(l_\alpha, \frac{1}{v_\alpha} \right) \right]. \end{aligned}$$

Solving for $V_H = 1/\langle 1/v_\alpha \rangle$ we get

Use the definition of the occupancy to derive an estimate of the average speed:

$$\boxed{V_H = \frac{Q \langle l_\alpha \rangle}{O \left[1 - \frac{Q}{O} \mathrm{Cov}(l_\alpha, 1/v_\alpha) \right]}.}$$

Chapter 5 :
Spatiotemporal Reconstruction of the Traffic State

Determining Speed from Single-Loop Detectors

Objective of this chapter:
Infer (reconstruct) the complete traffic state of an entire road—or even an entire road network—at any time and location, using limited and discrete traffic data, and do **spatiotemporal interpolation** (representing temporal and spatial "sampling"). Formulated in terms of a **convolution integral**.

Introduced a more refined interpolation method. This adaptive smoothing method yields a detailed and plausible reconstruction of the traffic state.

Combination and weighting of multiple, heterogeneous data sources for estimating the traffic state

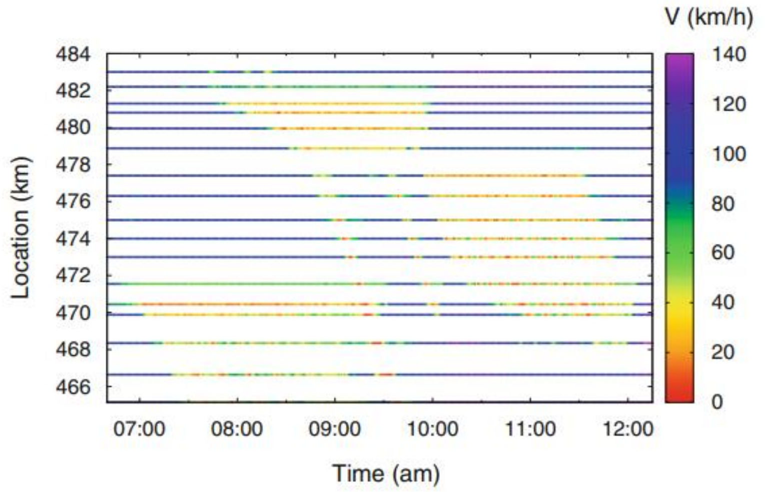
Spatiotemporal Interpolation

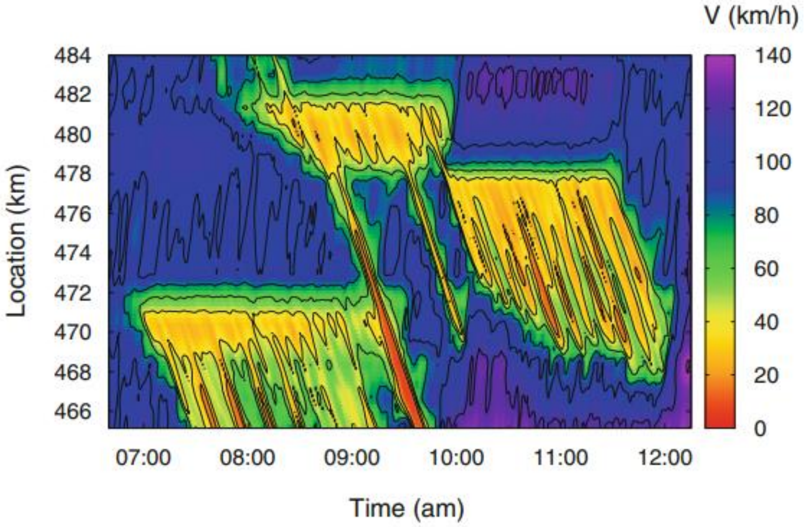
The purpose of the two-dimensional spatiotemporal interpolation algorithm described is to estimate the speed field

Input: Discrete data point $(x_{\mathrm{i}}, t_{\mathrm{i}}, v_{\mathrm{i}})$

Output: The output is the traffic-state estimator in the form of a continuous speed field (and possibly other fields) as a function of space and time.

Data are available in the form of aggregated Minute-by-minute data of speed and flow recorded by stationary detectors. Furthermore, floating cars transmitting “data telegrams” of their positions and time-mean speeds become increasingly relevant





Spatiotemporal Interpolation

The basis of spatiotemporal interpolation and smoothing is a discrete convolution with a kernel φ_0 that includes all data points i:

$$V(x,t) = \frac{1}{N(x,t)} \sum_i \phi_0(x - x_i, t - t_i) v_i$$

For Kernel,

- It should be localized, i.e., $\phi_0(x, t)$ tends to zero for sufficiently large values of $|x|$ and $|t|$.
- The maximum of $\phi_0(x, t)$ should be at $x = 0$ and $t = 0$.
- $\phi_0(x, t)$ should be a continuous function of x and t.
- $\phi_0(x, t)$ should be monotonically decreasing with $|x|$ and $|t|$.

The denominator N of denotes the normalization of the weighting function

Spatiotemporal Interpolation

For our purposes, the symmetric exponential has proved itself useful:

$$\phi_0(x - x_i, t - t_i) = \exp\left[-\left(\frac{|x - x_i|}{\sigma} + \frac{|t - t_i|}{\tau}\right)\right]$$

σ and τ are the smoothing widths in the spatial and temporal coordinates.

The exponential function operates as a low-pass filter, smoothing temporal variations on a scale smaller than τ and spatial fluctuations on a scale smaller than σ.

$$V(x,t) = \frac{1}{N(x,t)} \sum_i \phi_0(x - x_i, t - t_i) v_i$$

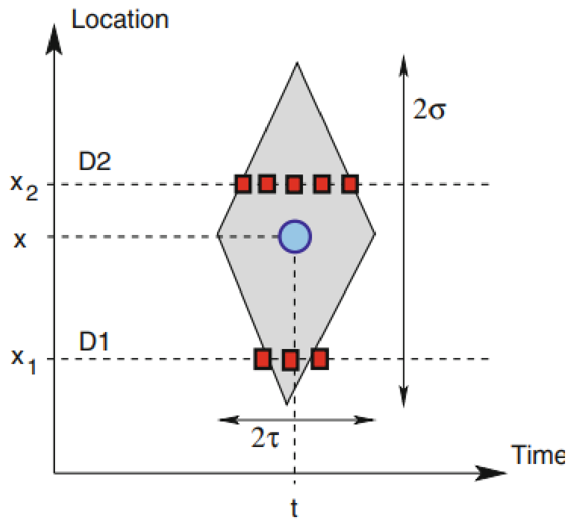
The normalization of the weighting function, given by the sum of all discrete weights:

$$\mathcal{N}(x, t) = \sum_i \phi_0(x - x_i, t - t_i)$$

Spatiotemporal Interpolation

$$V(x,t) = \frac{\sum_i \phi_0(x - x_i, t - t_i) v_i}{\sum_i \phi_0(x - x_i, t - t_i)}$$

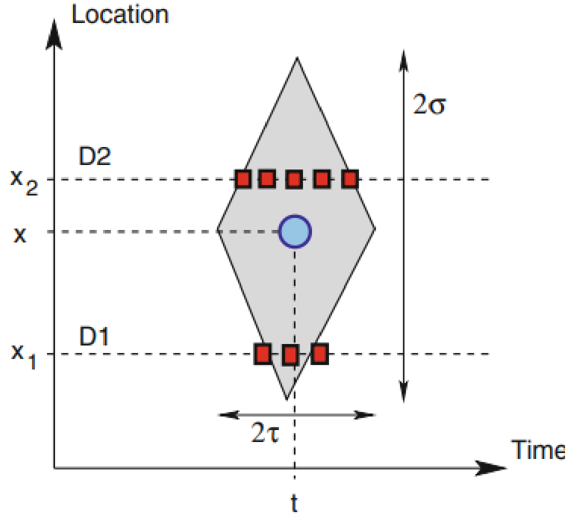
Limitations ; If the distance between two detectors is larger than half of the spatial distance between two stop-and-go waves, the **isotropic kernel** produces artifacts such as the “egg-carton pattern”. These introduce ambiguities into the interpretation of stop-and-go waves。

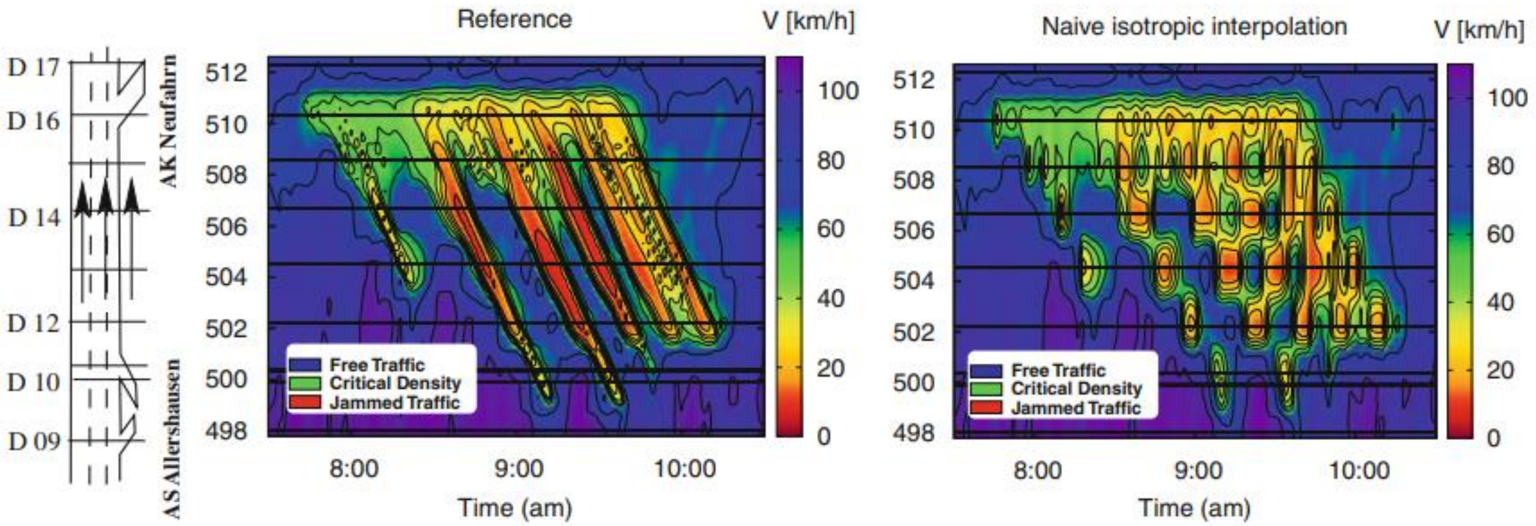


Spatiotemporal Interpolation

$$V(x,t) = \frac{\sum_i \phi_0(x - x_i, t - t_i) v_i}{\sum_i \phi_0(x - x_i, t - t_i)}$$

Limitations ; If the distance between two detectors is larger than half of the spatial distance between two stop-and-go waves, the **isotropic kernel** produces artifacts such as the “egg-carton pattern”. These introduce ambiguities into the interpretation of stop-and-go waves。

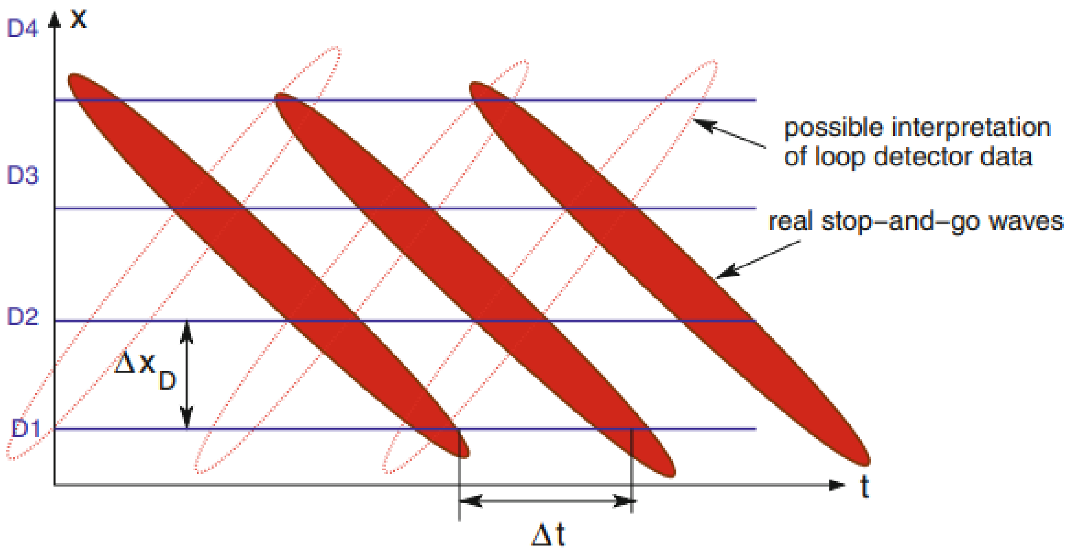




Spatiotemporal Interpolation

$$V(x,t) = \frac{\sum_i \phi_0(x - x_i, t - t_i) v_i}{\sum_i \phi_0(x - x_i, t - t_i)}$$

Limitations : If the distance between two detectors is larger than half of the spatial distance between two stop-and-go waves, the **isotropic kernel** produces artifacts such as the "egg-carton pattern". These introduce ambiguities into the interpretation of stop-and-go waves。



Adaptive Smoothing Method, ASM

Like the isotropic smoothing procedure, the adaptive smoothing method is based on a two-dimensional interpolation in space and time. In contrast to the former, it takes into account the **two typical velocities** of information propagation in **free** and **congested** traffic.

- First, the spatiotemporally averaged speeds (and other macroscopic quantities) are calculated using two different weighting kernels accounting for the different propagation velocities of macroscopic density and speed changes in free and congested traffic.
- Then, the two filters are used to determine the traffic state at the spatiotemporal location (x, t) by calculating the “degree of congestion”, w, which can take values between 0 and 1. The final speed field V(x, t) is a superposition of the two filters, weighted by *w*

ASM - Characteristic Propagation Velocities

Adaptive Smoothing Method takes into account that all perturbations to the traffic flow :

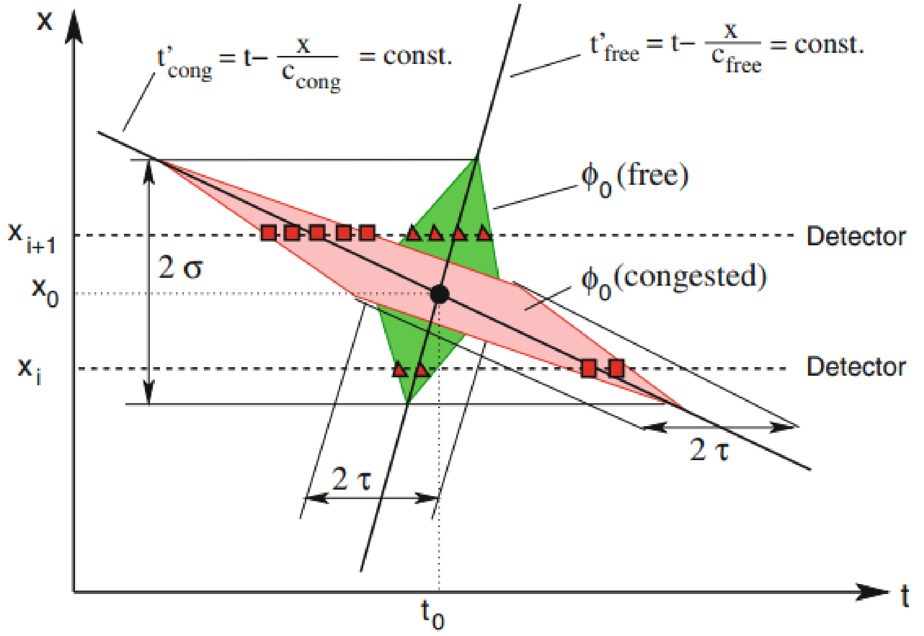
1. **Stationary**
2. **Free flow**: it has been observed that perturbations usually propagate with the traffic flow (downstream) at a characteristic velocity slightly below the local speed of the vehicles. This is due to the weak interactions between the vehicles.
3. **Congested traffic**: perturbations propagate against the traffic flow (upstream) due to the reaction of the drivers to their respective leading vehicles. Empirical data shows that the propagation velocity of approximately $c_{cong} = -15$ km/h is universal in congested traffic situations

ASM - Characteristic Propagation Velocities

To account for these fundamental properties of the traffics dynamics, two smoothed speed fields with different propagation velocities in free and congested traffic, c_{free} and c_{cong}, are considered

$$V_{\text{free}}(x,t) = \frac{1}{\mathcal{N}_{\text{free}}(x,t)} \sum_i \phi_0\left(x - x_i, t - t_i - \frac{x - x_i}{c_{\text{free}}}\right) v_i,$$

$$V_{\text{cong}}(x,t) = \frac{1}{\mathcal{N}_{\text{cong}}(x,t)} \sum_i \phi_0\left(x - x_i, t - t_i - \frac{x - x_i}{c_{\text{cong}}}\right) v_i$$



The normalization constants Nfree and Ncong are determined analogously to

$$\mathcal{N}(x,t) = \sum_i \phi_0(x - x_i, t - t_i)$$

ASM - Nonlinear Adaptive Speed Filter

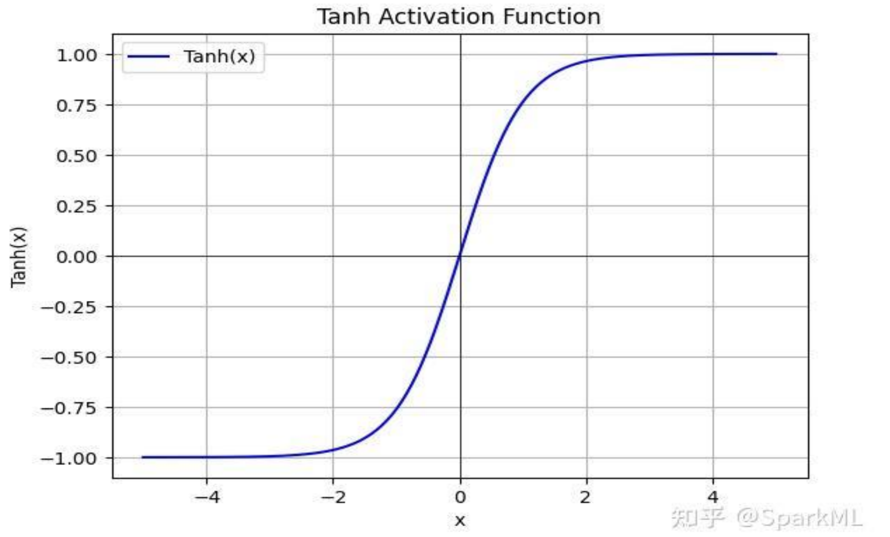
The result of the adaptive smoothing method—the average speed V (x, t)—is a superposition of the two speed fields $\boldsymbol{V_{free}}$ and $\boldsymbol{V_{cong}}$

$$V(x,t) = w(x,t)\, V_{\text{cong}}(x,t) + [1 - w(x,t)]\, V_{\text{free}}(x,t)$$

The weight w(x, t) depends on both Vfree and Vcong. Obviously, we want w ≈ 1 for low speeds and w ≈ 0 for high speeds. The continuous transition between the two extremes is characterized by an s-shaped (sigmoid) nonlinear function:

$$w(x,t) = \frac{1}{2}\left[1 + \tanh\left(\frac{V_c - V^*}{\Delta V}\right)\right]$$

The predictor $\boldsymbol{V^*(x,t)}$=min($\boldsymbol{V_{free}, V_{cong}}$) is defined such that congested traffic states are represented more accurately than free traffic. The parameter ΔV determines the transition width around V_c, which is the threshold between free and congested traffic. Good parameters values are, for example, V_c, = 55 km/h and ΔV = 10 km/h.



ASM - Paramaters

The table summarizes the six parameters of the adaptive smoothing method (ASM) and suggests suitable values for highway traffic.

ASM is very robust in the sense that the resulting speed field is insensitive to the precise values as long as the order of magnitude is correct

Parameter	Value (highway traffic)
Spatial smoothing width σ	$\Delta x/2$ (of the order of 1 km)
Temporal smoothing width τ	$\Delta t/2$ (of the order of 30 s)
Propagation velocity of perturbations in free traffic c_{free}	70 km/h
Propagation velocity of perturbations in congested traffic c_{cong}	−15 km/h
Threshold between free and congested traffic V_c	50 km/h
Width of the transition between free and congested traffic ΔV	10 km/h

ASM - Paramaters

The table summarizes the six parameters of the adaptive smoothing method (ASM) and suggests suitable values for highway traffic.

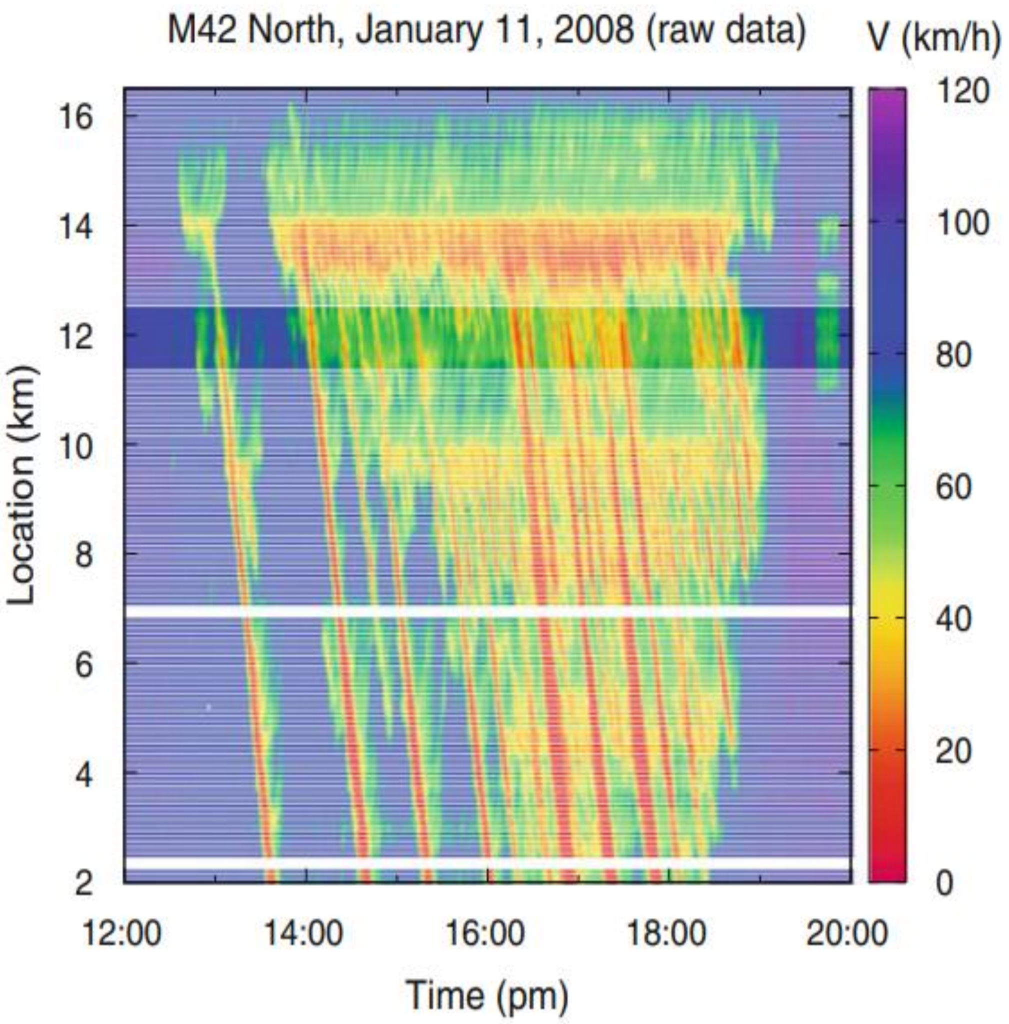
ASM is very robust in the sense that the resulting speed field is insensitive to the precise values as long as the order of magnitude is correct

Parameter	Value (highway traffic)
Spatial smoothing width σ	$\Delta x/2$ (of the order of 1 km)
Temporal smoothing width τ	$\Delta t/2$ (of the order of 30 s)
Propagation velocity of perturbations in free traffic c_{free}	70 km/h
Propagation velocity of perturbations in congested traffic c_{cong}	−15 km/h
Threshold between free and congested traffic V_c	50 km/h
Width of the transition between free and congested traffic ΔV	10 km/h

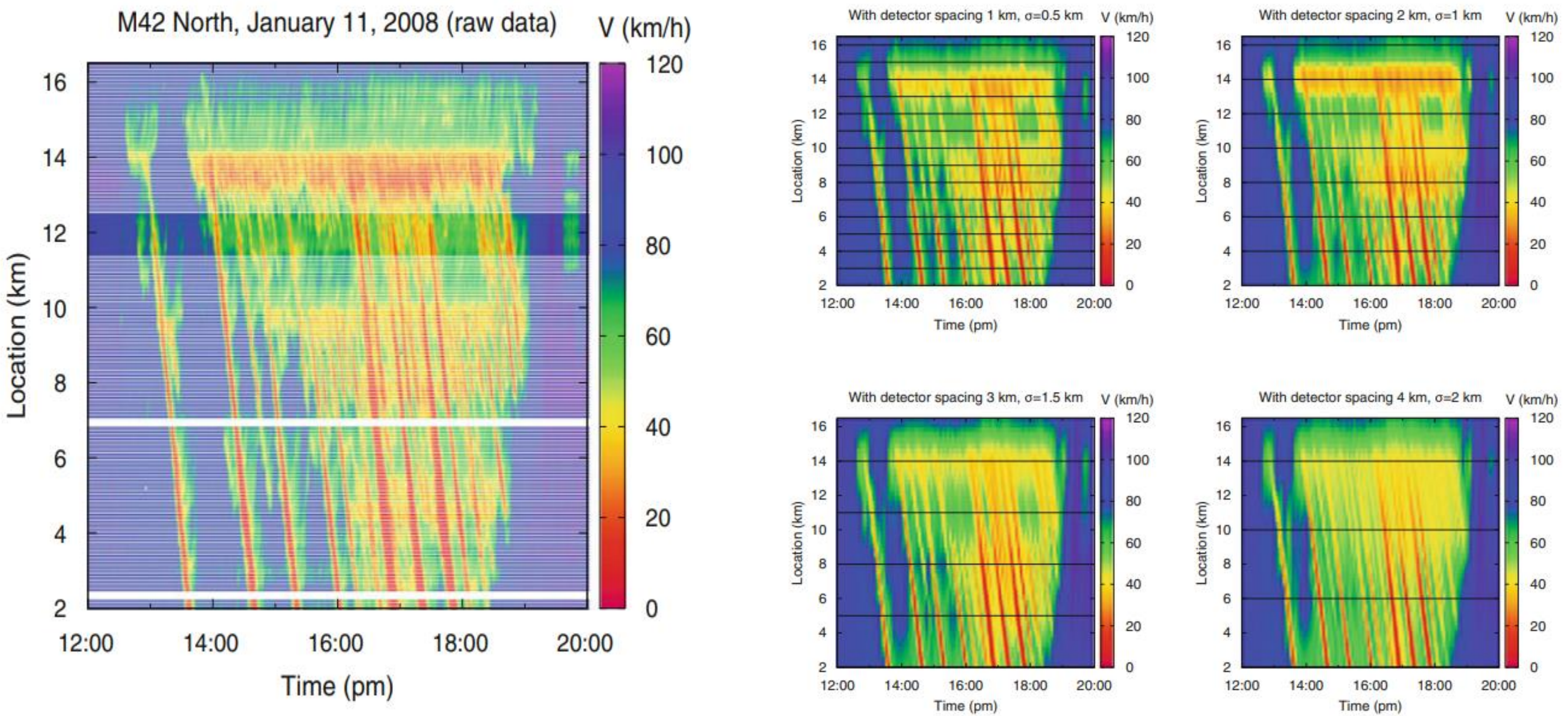
ASM - Testing the Predictive Power: Validation

Traffic-state recognition methods can be validated by applying them to a subset of given detector data and comparing the results with the full dataset. Ideally, the full data set serving as reference is so dense that it can be regarded as representing the ground truth .

The validation procedure consists in applying the ASM with standard parameters to input data chosen from just a small selection of the available detectors. The interpolated speed field V(x, t) is then compared to speed data at detectors which are half way between those whose data has been used in the reconstruction

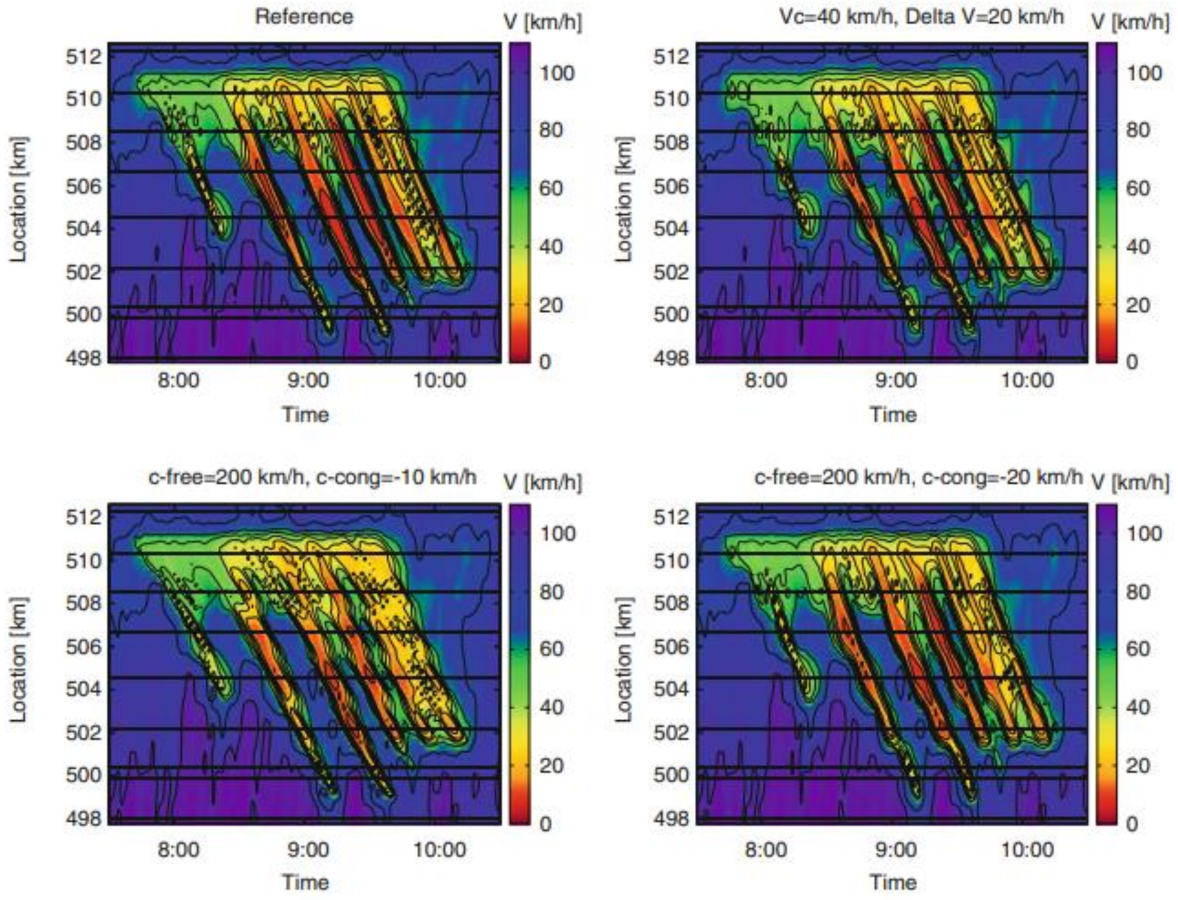


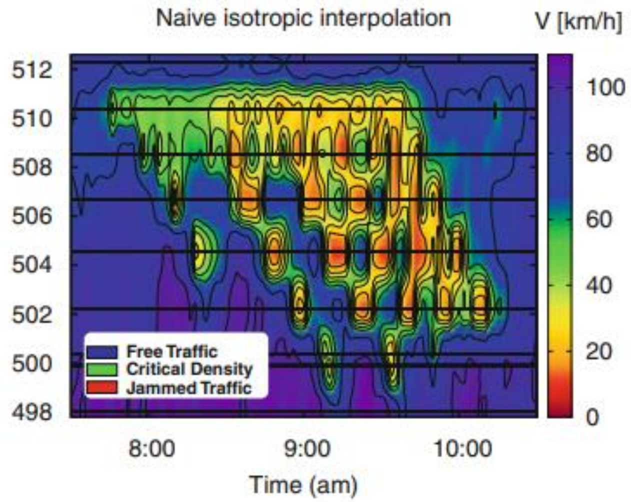
ASM - Testing the Predictive Power: Validation



ASM - Testing the Robustness: Sensitivity Analysis

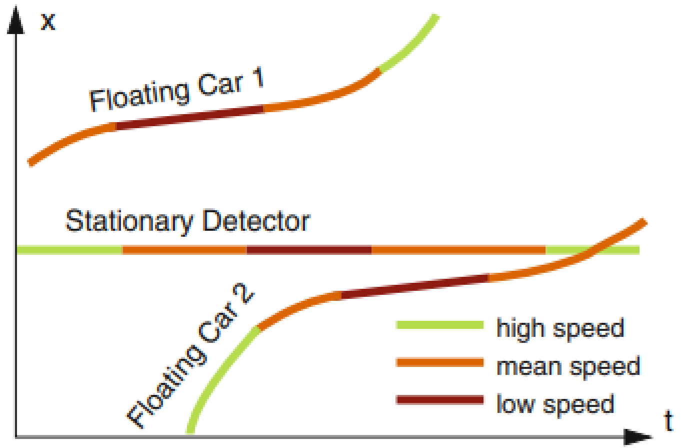
By varying the parameters listed we can check the "robustness" of the traffic state reconstruction, i.e., the sensitivity of the result to changes in the parameter values.





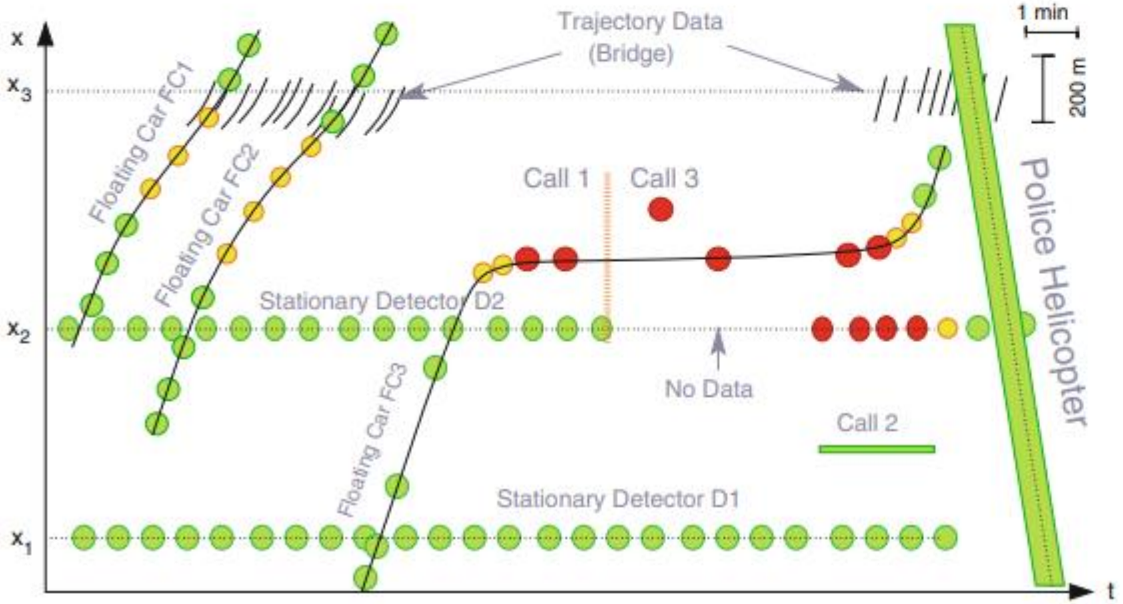
Data Fusion

The term data fusion refers to the process of combining data from multiple, heterogeneous data sources such as **cross-sectional data, floating-car data, "floating-phone data", police reports**, etc. In general, each of these categories of data describes different aspects of the traffic situation and might even contradict each other. The goal of data fusion is to maximize the utility of the available information



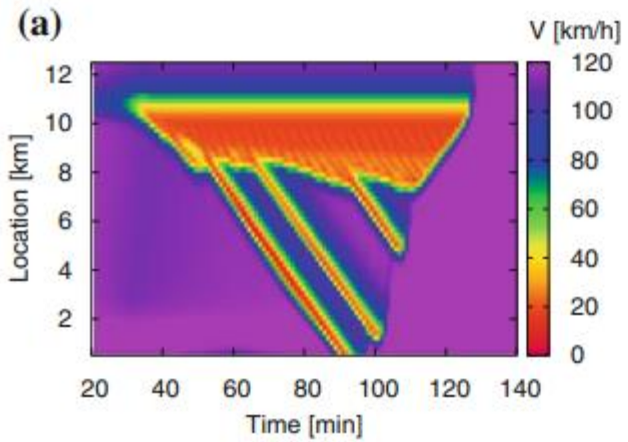
Data Fusion

The term data fusion refers to the process of combining data from multiple, heterogeneous data sources such as **cross-sectional data, floating-car data, “floating-phone data”, police reports**, etc. In general, each of these categories of data describes different aspects of the traffic situation and might even contradict each other. The goal of data fusion is to maximize the utility of the available information



Data Fusion

The adaptive smoothing method (ASM) can be used as an algorithm for data fusion if all data sources provide spatiotemporally resolved point measurements of the local speed (i.e., data sets $\{x_i, t_i, v_i\}$)



This includes stationary detector data (SDD) and floating-car data (FCD).

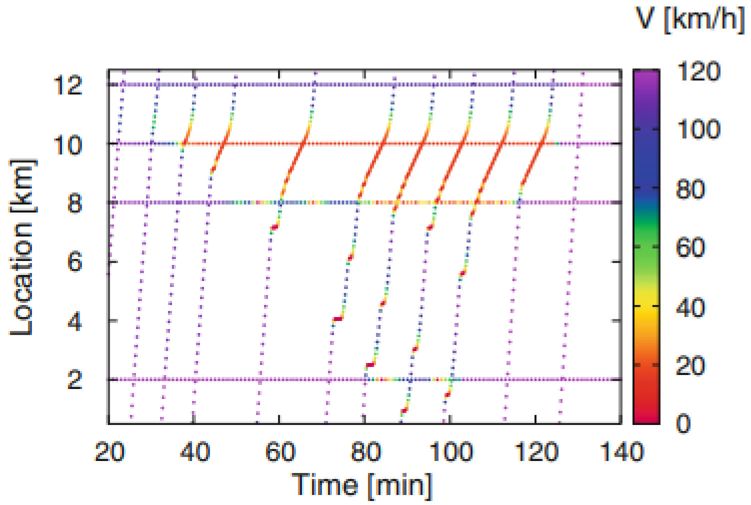
As empirical test cases containing simultaneous SDD, FCD, and ground-truth data are extremely rare, we demonstrate how to validate data fusion procedures using model-based simulation. To this end, we simulate traffic waves using a human driver model capable of realistically reproducing them.

Data Fusion

The adaptive smoothing method (ASM) can be used as an algorithm for data fusion if all data sources provide spatiotemporally resolved point measurements of the local speed (i.e., data sets $\{x_i, t_i, v_i\}$)

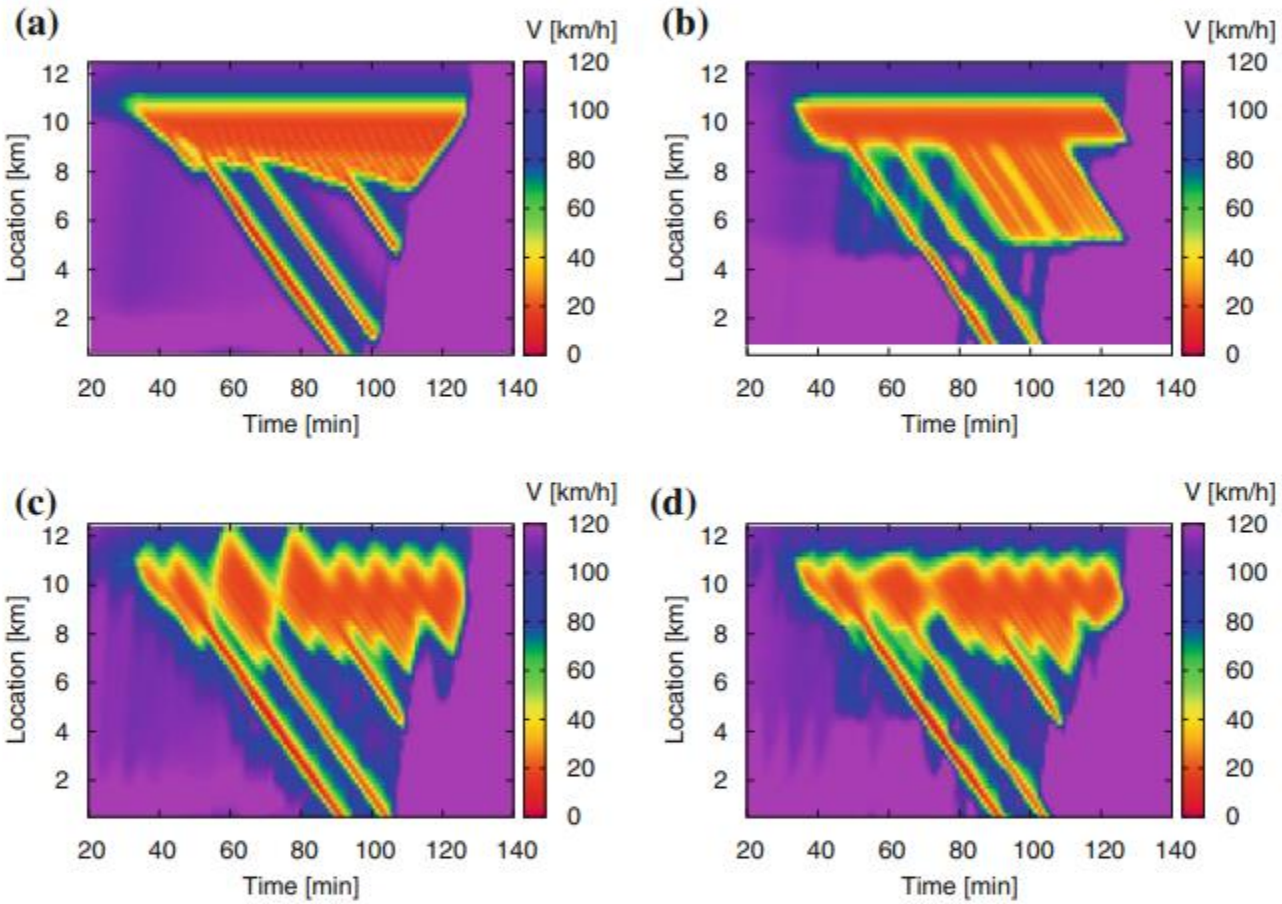
This includes stationary detector data (SDD) and floating-car data (FCD).

As input for the adaptive smoothing method, we generate virtual SDD and FCD from the simulation and apply the method with the standard parameters.



Data Fusion

a) Ground truth
b) Using stationary detector data (SDD) only
c) Using floating-car data (FCD) only
d) Combining stationary detector and floating-car data.



Weighting the data source

When using multiple data sources, their relative weighting plays an important role.

However, In AMI, the weight is:

$$w_i(x,t) = \phi_0\left(x - x_i,\ t - t_i - \frac{x - x_i}{c}\right)$$

depend only on the (spatiotemporal) distance between the location in question (x, t) and the data points (x_i, t_i). Consequently, all data points are considered equally important.

If the data originate from several detector categories m (such as induction-loop detectors, infrared detectors, floating cars) with different magnitude of the associated errors, it is sensible to include an additional weight r_m that represents the reliability of the data source to give the more reliable sources a stronger influence on the result.

$$w_i^{tot}(x,t) = w_i(x,t) r_{m(i)} = \phi_0\left(x - x_i,\ t - t_i - \frac{x - x_i}{c}\right) r_{m(i)}.$$

m(i) denotes that the data source i is of type m with reliability weight r_m

Assume that

(i) The different data sources m bear no systematic errors

(ii) The variance θ_m of the random errors is known

(iii) The errors of the different sources are uncorrelated.

Weighting the data source

$$w_i^{tot}(x,t) = w_i(x,t) r_{m(i)} = \phi_0\left(x - x_i,\ t - t_i - \frac{x - x_i}{c}\right) r_{m(i)}.$$

Assume that, for a given point (x, t), speed estimates V_m from all data types are available, such that

$$\phi_0(x - x_i, t - t_i - (x - x_i)/c) = 1$$

Then, according to a basic addition rule for a linear combination of independent random variables, the error variance of the weighted arithmetic mean $V(x,t) = \sum_m r_m v_m$ is given by $\theta = \sum_m r_m^2 \theta_m$,where $\sum_m r_m = 1$ must be satisfied

$$\theta = \mathrm{Var}(V) = \mathrm{Var}\left(\sum_m r_m v_m\right) = \sum_m r_m^2 \mathrm{Var}(v_m) = \sum_m r_m^2 \theta_m$$

Our objective is to minimize this variance by varying the weights r_m , or the weight vector $\mathbf{r}$. This immediately leads to following constrained optimization problem:

Minimize $$\theta(\mathbf{r}) = \sum_m r_m^2 \theta_m,$$

S.T. $$\sum_m r_m = 1.$$

Weighting the data source

Minimize $$\theta(\mathbf{r}) = \sum_m r_m^2 \theta_m ,$$

S.T. $$\sum_m r_m = 1 .$$

拉格朗日乘子法解决问题 Lagrange Multipliers.

① 约束函数构建: $B_1(r) = \sum_m r_m - 1$.

②. Define Lagrange function

$$\mathcal{L}(r, \lambda) = \theta r - \sum_n \lambda_n B_n(r) .$$

$$L(r, \lambda_1) = \sum_m r_m^2 \theta_m - \lambda_1 \left(\sum_m r_m - 1 \right)$$

$$L(r_1 \cdots r_m \lambda_1) = r_1^2 \theta_1 + r_2^2 \theta_2 + \cdots r_m^2 \theta_m - \lambda_1 \left(\sum_m r_m - 1 \right).$$

③. 极小值必要条件: 拉格朗日求偏导

$$\frac{\partial L(r, \lambda)}{\partial r} = 0 .$$

m个方程:

$$\frac{\partial L}{\partial r_m} = 2 r_m \theta_m - \lambda_1$$

令其为0，则 $r_m = \frac{\lambda_1}{2\theta_m}$

Weighting the data source

Minimize $\theta(\mathbf{r}) = \sum_m r_m^2 \theta_m ,$

S.T. $\sum_m r_m = 1 \, .$

Conclusion: The weights should be proportional to the reciprocal of the error variance in the data source.

Total spatiotemporal weight in the final fusion formula:

$$w_i^{\text{tot}}(x,t) = \phi_0 \left(x - x_i ,\, t - t_i - \frac{x - x_i}{c} \right) \cdot \frac{\frac{1}{\theta_{m(i)}}}{\sum_j \frac{1}{\theta_j}}$$

4. 求解未知 拉式乘子

$\cancel{r_m^2 \theta_m}$ ∵ 有个约束: $\sum_m r_m = 1$

$$\sum_m \frac{\lambda}{2\theta_m} = 1$$

$$\frac{\lambda}{2} \sum_m \frac{1}{\theta_m} = 1$$

$$\frac{r_m}{\theta_m} \quad \frac{\lambda}{2} = \frac{1}{\sum_m \frac{1}{\theta_m}}$$

$$r_m = \frac{\frac{1}{\theta_m}}{\sum_{m'} \frac{1}{\theta_{m'}}}$$

权重∝ Inverse of variance of errors